A Tight Integrality Gap for the Time-Indexed LP for Bipartite AND/OR Scheduling

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Abstract

Happach and Schulz introduced a time-indexed linear programming relaxation for single-machine scheduling with bipartite AND/OR precedence constraints and proved a 2Δ approximation guarantee by randomized α-point rounding, where Δ is the maximum number of OR-predecessors of a job. They asked whether the Section 2 analysis is tight, equivalently whether stronger bounds are possible for the integrality gap of the same time-indexed formulation. We show that the integrality gap is exactly 2Δ in the supremal sense. The lower bound is a complete Δ-uniform hitting-set gadget: fractionally, all Δ-sets are hit at rate Δ/n, while integrally one must choose $n - \Delta + 1$ unit jobs before a terminal zero-processing job can complete.

1 Introduction

Happach and Schulz [1] study the single-machine problem

$$1 \mid \textit{ao-prec} = A \dot{\cup} B,\ p_j \in \{0, 1\} \mid \sum_j w_j C_j,$$

where the precedence constraints consist of AND-arcs inside A or inside B and OR-arcs from A to B. In their Section 2, they give a time-indexed linear programming relaxation and a randomized α-point rounding algorithm with performance guarantee 2Δ, where

$$\Delta := \max_{b \in B} |P(b)|$$

and $P(b)$ denotes the set of OR-predecessors of b. In the conclusion, they raise the question whether the analyses of the Section 2 algorithms are tight and, in particular, whether one can obtain stronger bounds on the integrality gap of the Section 2 time-indexed formulation.

This note gives a matching lower bound. The example is deliberately simple. Let A consist of $n = m\Delta$ unit jobs. For every Δ-subset $S \subseteq A$ we create a zero-processing job b_S whose OR-predecessor set is exactly S, and we add one zero-processing terminal job z that AND-depends on every b_S. The LP can spread the unit jobs of A evenly over the horizon, so every Δ-set is fractionally hit by time about n/Δ. But any integral schedule must hit every Δ-subset of A before completing z, which requires processing $n - \Delta + 1$ of the unit jobs. This gives ratio tending to 2Δ.

Together with the upper bound of Happach and Schulz, the result is the following.

Theorem 1. *For every integer $\Delta \geq 1$, the supremal integrality gap of the Section 2 time-indexed LP of Happach and Schulz for instances with $\max_{b\in B} |P(b)| \leq \Delta$ is exactly*

$$2\Delta.$$

Moreover, the lower-bound instances have $p_j \in \{0,1\}$, weights in $\{0,1\}$, and AND-arcs only inside B.

2 The time-indexed relaxation

We recall the relaxation in the normalized form used by Happach and Schulz. The job set is partitioned as $N = A \dot{\cup} B$. Each job has processing time $p_j \in \{0,1\}$. As in [1], weights on jobs in A may be shifted to zero-processing successors in B, so we take $w_a = 0$ for $a \in A$. Let

$$T := \sum_{j\in N} p_j$$

be the time horizon and write $[T] = \{1,\dots,T\}$. The variable x_{jt} denotes the fractional event that job j completes at time t. For convenience, define

$$F_j(t) := \sum_{s=1}^{t} x_{js}, \qquad F_j(0) := 0.$$

The relaxation is

$$\begin{aligned}
\min \quad & \sum_{b\in B} w_b \sum_{t=1}^{T} t x_{bt} && && (1)\\
\text{s.t.} \quad & \sum_{t=1}^{T} x_{jt} = 1 && \forall j \in N, && (2)\\
& \sum_{j\in N} \sum_{s=t-p_j+1}^{t} x_{js} \leq 1 && \forall t \in [T], && (3)\\
& F_b(t+p_b) \leq \sum_{a\in P(b)} F_a(t) && \forall b \in B : P(b) \neq \emptyset,\ \forall t \in \{1,\dots,T-p_b\}, && (4)\\
& F_j(t+p_j) \leq F_i(t) && \forall (i,j) \in E_\wedge,\ \forall t \in \{1,\dots,T-p_j\}, && (5)\\
& x_{jt} \geq 0 && \forall j \in N,\ \forall t \in [T]. && (6)
\end{aligned}$$

Empty sums in (3) are interpreted as zero; hence zero-processing jobs do not consume capacity. Happach and Schulz prove via randomized α-point rounding that every feasible fractional solution can be rounded to a feasible schedule of expected objective value at most 2Δ times the LP objective value. Consequently,

$$\mathrm{OPT}_{\mathrm{IP}}(I) \leq 2\Delta\ \mathrm{OPT}_{\mathrm{LP}}(I) \qquad (7)$$

for every instance I of the above form. We now prove that the factor in (7) cannot be improved.

3 The lower-bound construction

Fix an integer $\Delta \geq 1$ and a positive integer m. Let

$$n := m\Delta.$$

We construct an instance $I_{m,\Delta}$ as follows. Let

$$A = \{a_1, \ldots, a_n\}, \qquad p_a = 1, \qquad w_a = 0 \quad \text{for all } a \in A.$$

For every Δ-subset $S \subseteq A$, introduce a job $b_S \in B$ with

$$p_{b_S} = 0, \qquad w_{b_S} = 0, \qquad P(b_S) = S.$$

Finally introduce a terminal job $z \in B$ with

$$p_z = 0, \qquad w_z = 1, \qquad P(z) = \emptyset.$$

For every $S \in \binom{A}{\Delta}$, add the AND-precedence arc

$$b_S \prec z.$$

There are no other precedence constraints. Thus the maximum number of OR-predecessors of any job in B is exactly Δ.

3.1 The integer optimum

Lemma 1. *For the instance $I_{m,\Delta}$,*

$$\mathrm{OPT}_{\mathrm{IP}}(I_{m,\Delta}) = n - \Delta + 1.$$

Proof. Consider any feasible schedule and let

$$H := \{a \in A : C_a \leq C_z\}$$

be the set of unit jobs from A completed no later than the terminal job z. For each Δ-set $S \subseteq A$, the AND-arc $b_S \prec z$ gives $C_z \geq C_{b_S}$, and the OR-constraint for b_S gives

$$C_{b_S} \geq \min_{a \in S} C_a$$

because $p_{b_S} = 0$. Hence, for every $S \in \binom{A}{\Delta}$, at least one job of S belongs to H. Thus H is a hitting set for the complete Δ-uniform hypergraph on A.

A subset $H \subseteq A$ hits every Δ-subset of A if and only if its complement has size at most $\Delta - 1$. Therefore

$$|H| \geq n - \Delta + 1.$$

Since all jobs in A have processing time 1, completing all jobs in H no later than z requires

$$C_z \geq |H| \geq n - \Delta + 1.$$

The terminal job z is the only job of positive weight, so every feasible schedule has objective value at least $n - \Delta + 1$.

Conversely, choose any set $H \subseteq A$ with $|H| = n - \Delta + 1$ and schedule first all jobs in H. Since $|A \setminus H| = \Delta - 1$, every Δ-subset $S \subseteq A$ intersects H. Hence, after the jobs in H have completed, every job b_S has at least one completed OR-predecessor. We may then schedule all zero-processing jobs b_S and then z, all at completion time $n - \Delta + 1$, and schedule the remaining jobs of A afterwards. This schedule is feasible and has objective value $n - \Delta + 1$. $\square$

3.2 The LP optimum

Lemma 2. *For the instance $I_{m,\Delta}$,*

$$\mathrm{OPT}_{\mathrm{LP}}(I_{m,\Delta}) = \frac{m+1}{2}.$$

Proof. The time horizon is $T = n$. We first prove the lower bound. Let x be any feasible solution of the LP. For every $t \in \{0, 1, \ldots, n\}$ and every $S \in \binom{A}{\Delta}$, the AND-constraint $b_S \prec z$ and $p_z = 0$ imply

$$F_z(t) \le F_{b_S}(t),$$

and the OR-constraint for b_S, together with $p_{b_S} = 0$, implies

$$F_{b_S}(t) \le \sum_{a \in S} F_a(t).$$

Therefore

$$F_z(t) \le \min_{S \in \binom{A}{\Delta}} \sum_{a \in S} F_a(t).$$

The minimum is at most the average over all Δ-subsets of A, so

$$\begin{aligned} F_z(t) &\le \frac{1}{\binom{n}{\Delta}} \sum_{S \in \binom{A}{\Delta}} \sum_{a \in S} F_a(t) \\ &= \frac{\Delta}{n} \sum_{a \in A} F_a(t). \end{aligned}$$

By summing the capacity inequalities (3) over the first t time slots, using $p_a = 1$ for $a \in A$ and $p_b = 0$ for $b \in B$, we get

$$\sum_{a \in A} F_a(t) \le t.$$

Consequently

$$F_z(t) \le \min\left\{\frac{\Delta t}{n}, 1\right\} = \min\left\{\frac{t}{m}, 1\right\}. \tag{8}$$

We use the discrete tail identity

$$\sum_{t=1}^{n} t x_{zt} = \sum_{q=0}^{n-1} \big(1 - F_z(q)\big),$$

which follows by writing $t = \sum_{q=0}^{t-1} 1$ and exchanging the two sums. Applying (8),

$$\sum_{t=1}^{n} t x_{zt} \ge \sum_{q=0}^{m-1} \left(1 - \frac{q}{m}\right) = \frac{m+1}{2}.$$

Since z is the only positive-weight job, every feasible LP solution has objective value at least $(m+1)/2$.

It remains to exhibit a feasible solution attaining this value. For every $a \in A$, set

$$x_{a,t} = \frac{1}{n} \qquad (t = 1, \dots, n).$$

For every $S \in \binom{A}{\Delta}$, and for the terminal job z, set

$$x_{b_S,t} = x_{z,t} = \begin{cases} 1/m, & 1 \le t \le m, \\ 0, & m < t \le n. \end{cases}$$

The assignment constraints are immediate. At each time t, the only jobs that consume capacity are the jobs in A, and

$$\sum_{a \in A} x_{a,t} = n \cdot \frac{1}{n} = 1,$$

so the capacity constraints hold. For $S \in \binom{A}{\Delta}$ and every t,

$$F_{b_S}(t) = \min\left\{\frac{t}{m}, 1\right\} \le \frac{t}{m} = \frac{\Delta t}{n} = \sum_{a \in S} F_a(t),$$

so all OR-constraints hold. Moreover $F_z(t) = F_{b_S}(t)$ for every S and every t, so all AND-constraints $b_S \prec z$ hold. The LP objective value of this solution is

$$\sum_{t=1}^{n} t x_{zt} = \frac{1}{m} \sum_{t=1}^{m} t = \frac{m+1}{2}.$$

Thus the LP optimum is exactly $(m + 1)/2$. □

4 Proof of the gap theorem

Proof of Theorem 1. The upper bound is (7), due to Happach and Schulz. For the matching lower bound, apply Lemmas 1 and 2 to $I_{m,\Delta}$. The integrality gap of this instance is

$$\frac{\mathrm{OPT}_{\mathrm{IP}}(I_{m,\Delta})}{\mathrm{OPT}_{\mathrm{LP}}(I_{m,\Delta})} = \frac{n - \Delta + 1}{(m + 1)/2}.$$

Since $n = m\Delta$, this is

$$\frac{2(m\Delta - \Delta + 1)}{m + 1}.$$

Letting $m \to \infty$ gives

$$\lim_{m \to \infty} \frac{2(m\Delta - \Delta + 1)}{m + 1} = 2\Delta.$$

Hence, for every $\varepsilon > 0$, there is an instance with maximum OR-in-degree Δ whose LP gap is at least $2\Delta - \varepsilon$. Together with the upper bound 2Δ, this proves that the supremal integrality gap is exactly 2Δ. □

Remark 1. *The construction uses AND-arcs of the form $b_S \prec z$ inside B, which are allowed in the bipartite AND/OR model of Happach and Schulz. Thus it applies to the Section 2 LP for the stated AND/OR problem. It does not, by itself, prove tightness for the narrower subclass with $E_\wedge = \emptyset$.*

References

[1] F. Happach and A. S. Schulz. Approximation algorithms and linear programming relaxations for scheduling problems related to min-sum set cover. *Mathematics of Operations Research*, 49(1):578–598, 2024. doi:10.1287/moor.2023.1368.